CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

[redacted]

Lecture No.- 10

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Recap of Previous Lecture

Topic Connectivity in Graph Part-02

Topics to be Covered

Topic Connectivity in Graph Part-03

Topic : Connectivity in Graphs

$n = 10 \quad k = 2$

min no. of edges

$e = n - k$

$= 10 - 2$

$e = 8$

Case 1

Tree: $e = 8$ (9V) + Tree: $e = 0$ (1V) $= 8$

Case 2

Tree: $e = 7$ (8V) + Tree: $e = 1$ (2V) $= 8e$

Case 3

Tree: $e = 6$ (7V) + $e = 2$ (3V) $= 8$

$$\left\{ \begin{array}{ll} 9V & 1V \\ 8V & 2V \\ 7V & 3V \\ \vdots & \end{array} \right.$$

$n = 10 \quad k = 2$

$$e = \frac{(n-k)(n-k+1)}{2}$$

$$= \frac{(10-2)(10-2+1)}{2}$$

$$= \frac{\cancel{8}^{4} \cdot 9}{\cancel{2}}$$

$$= 36e$$

$K_9 \qquad K_1$

Case 1 ✗

$\frac{\cancel{6}^{3} \times 5}{\cancel{2}} + \frac{\cancel{4}^{2} \cdot 3}{\cancel{2}}$

$15\ K_6 + 6\ K_4 = 21e$

$6v \qquad 4v$

Case 2 ✗

$\frac{7 \times 6}{2} = 21e + \frac{3 \cdot 2}{2} = 3e = 24$

$7v \qquad 3v$

Case 3 ✗

$\frac{\cancel{8}^{4} \cdot 7}{\cancel{2}} = 28 + \frac{\cancel{2} \cdot 1}{\cancel{2}}\ 1e = 29e$

$8v \qquad 2v$

Case 4 ✓

$\frac{9 \cdot 8}{2} = 36e + 0e\ 0 = 36e$

$9v \qquad 1v$

$n = 11 \quad K = 3.$

$e = n - K.$

$= 11 - 3 = 8e$

$\underset{6V}{\overset{6-1}{= 5e}} + \underset{3V}{= 2e} + \underset{2V}{= 1e} = 8e.$

$\underset{5V}{4e} + \underset{3V}{2e} + \underset{3V}{2e} = 8e$

$n = 11 \quad k = 3.$

max no. of edges.

$$e = (n-k)(n-k+1)/2.$$

$$= (11-3)(11-3+1)/2$$

$$= \frac{8 \cdot 9}{2} = 36.$$

$$\underset{5V}{\left(\begin{matrix}5 \cdot 4/2 \\ = 10\end{matrix}\right)} + \underset{4V}{\left(\begin{matrix}4 \cdot 3/2 \\ = 6\end{matrix}\right)} + \underset{2V}{\left(\begin{matrix}2 \cdot 1/2. \\ = 1.\end{matrix}\right)} = 17 \times.$$

$$\underset{6V}{\left(\begin{matrix}6 \cdot 5/2 \\ = 15\end{matrix}\right)} + \underset{3V}{\left(\begin{matrix}3 \cdot 2/2 \\ = 3\end{matrix}\right)} + \underset{2V}{\left(\begin{matrix}2 \cdot 1/2 \\ = 1.\end{matrix}\right)} = 19 \times$$

$$\underset{1V}{(0e)} + \underset{1V}{(0e)} + \underset{9V}{\left(\begin{matrix}9 \cdot 8/2 \\ = 36.\end{matrix}\right)} = \underline{36e.}$$

Consider a Graph having 100 vertices & 4 component then max no of edges?

$n = 100 \quad k = 4$

$e = (n-k)(n-k+1)/2$

$= (100-4)(100-4+1)/2$

$= \frac{96 \cdot 97}{2}$

Consider a Graph having 100 vertices & 4 component

1st is having 25 vertices.

2nd — " — 24 vertices

3rd — " — 26 vertices

4th — " — 25 vertices

max no of edges?

25   26v   24v   25

$\frac{25 \cdot 24}{2} + \frac{26 \cdot 25}{2} + \frac{24 \cdot 23}{2} + \frac{25 \cdot 24}{2}$

n=100 k=4.

20 20 30 30

.
.
.

$\boxed{25 \quad 26 \quad 24 \quad 25}$ Q2.

.
.
.

$\boxed{1\% \quad 1\% \quad 1\% \quad 97\%}$ Q1.

$k \geq 2$.

Consider a $\boxed{\text{Disconnected Graph}}$ of 10 vertices. What will be max no. of edges it can have?

$k \uparrow \quad e \downarrow$

max

$k = 2$.

$n = 10$

$e = (n-k)(n-k+1)/2$.

$= (10-2)(10-2+1)/2$.

$= \frac{8 \cdot 9}{2}$

$= 36$.

$K_1 \quad K_9$

$k = 3$

$e = (10-3)(10-3+1)/2$.

$= \frac{7 \cdot 8}{2}$.

$= 24$.

$k = 4$.

$e = (10-4)(10-4+1)/2$.

$= \frac{\cancel{6}^{3} \cdot 7}{\cancel{2}} = 21 e$.

$k \geq 2$.

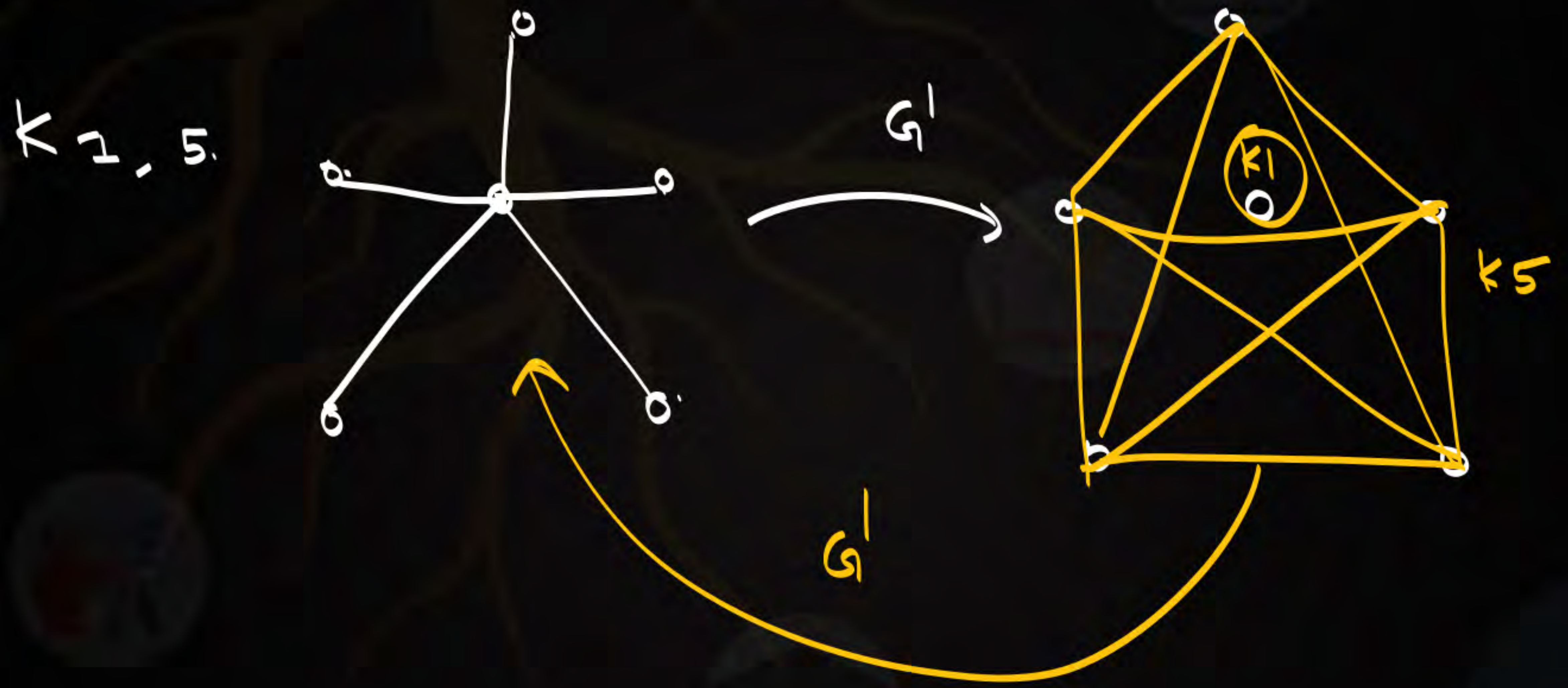
Consider a $\boxed{\text{Disconnected Graph}}$ of 10 vertices with 'max no. of edges' K_1 K_9

if we take complement of this graph what will be total edges?

K_1 K_9 $\xrightarrow{G'}$ Star Graph. $\xrightarrow{G'}$ K_9

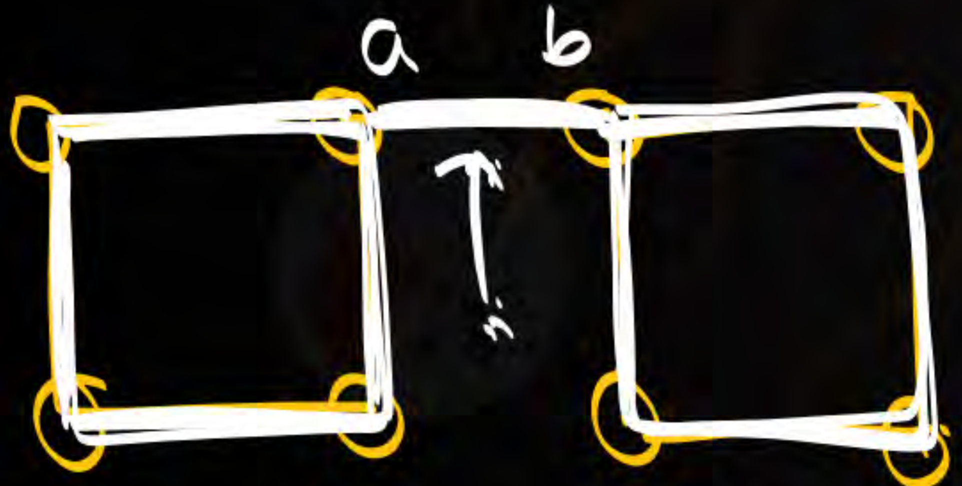
$K_{1,9}$

$e(G') = 9$.


K1, 5
G'
K1
K5
G'

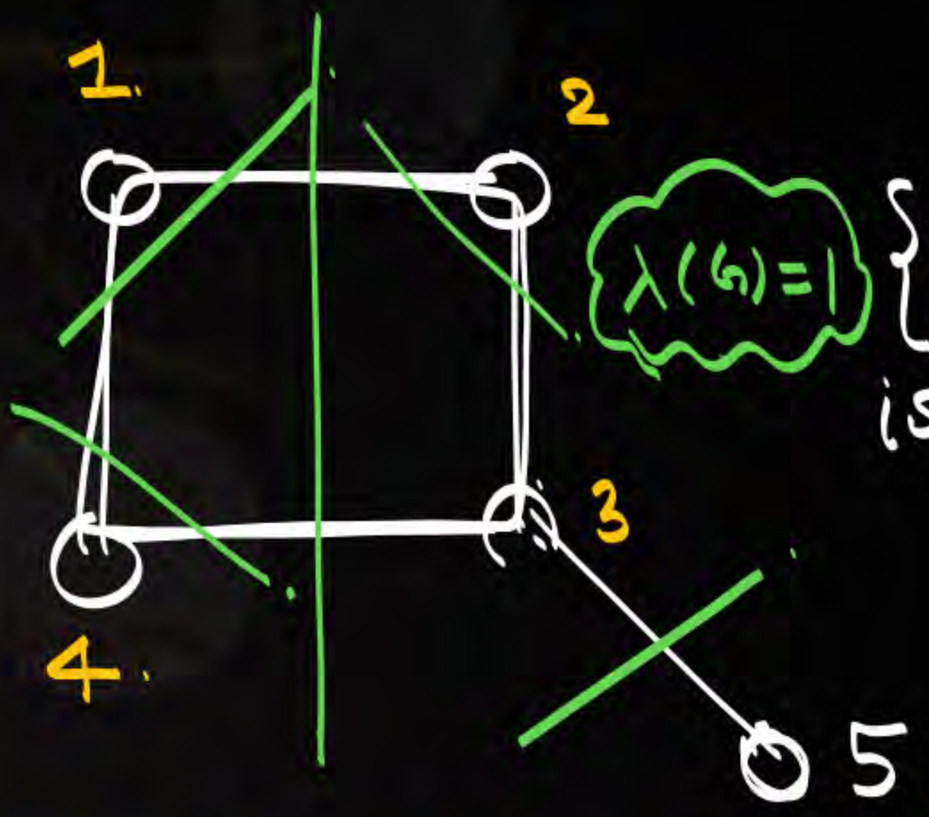
Cut edge / bridge :

Removal of $\boxed{\text{single edge}}$ from a Graph, will make graph as a disconnected Graph.



$\lambda(G) = 1$

Cut edge : {ab}



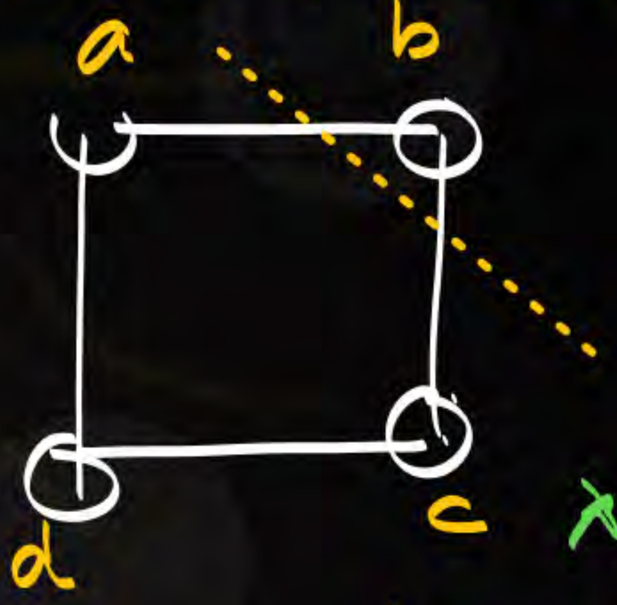
$\lambda(G) = 1$ {3 5} is cut edge

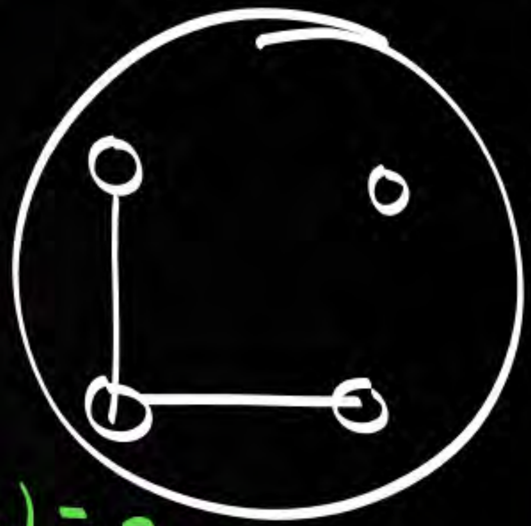
* if cut edge exist it does not belongs to cycle.

Cut edgeset / Cut set :

Removal of set of edges from a Graph will make graph as a disconnected Graph.

Cutset: {ab, bc}.

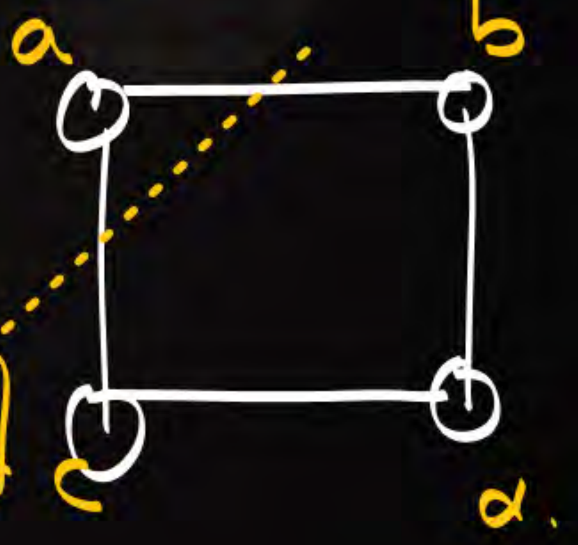


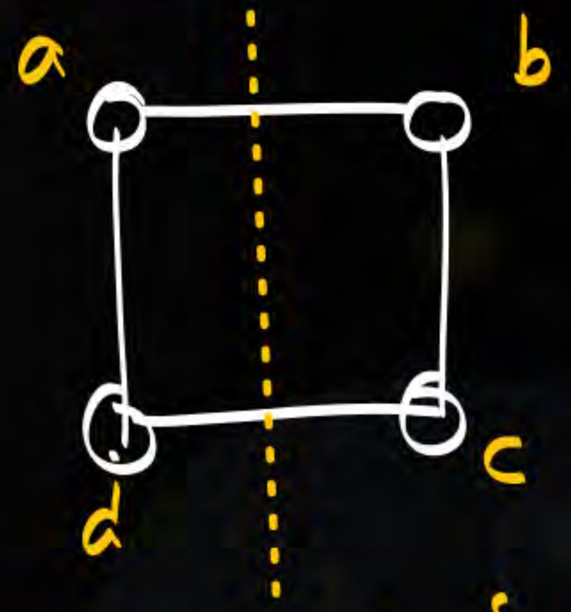


K=2

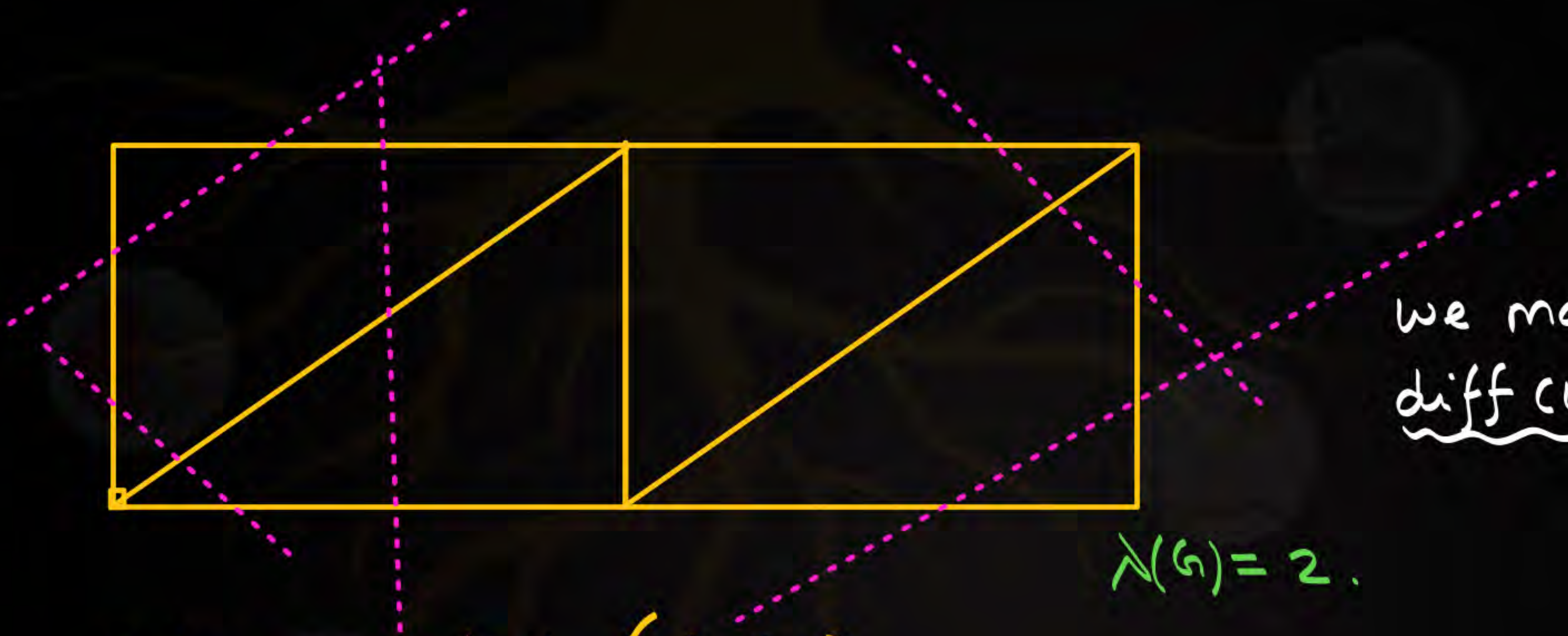
$\lambda(G) = 2$.

cutset : {ab, ac}





cutset: {ab, dc}



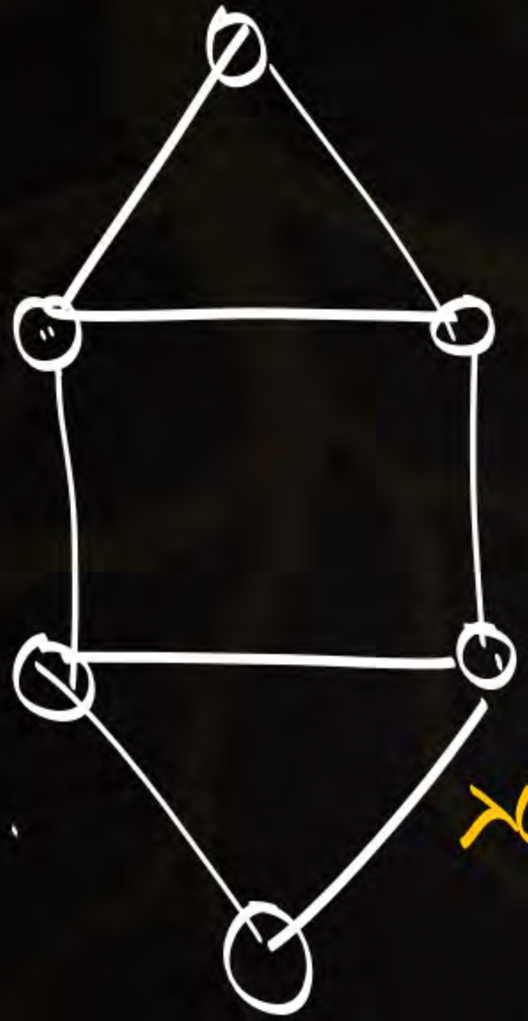
$\lambda(G) = 2.$

we may have diff cut set's of diff sizes.

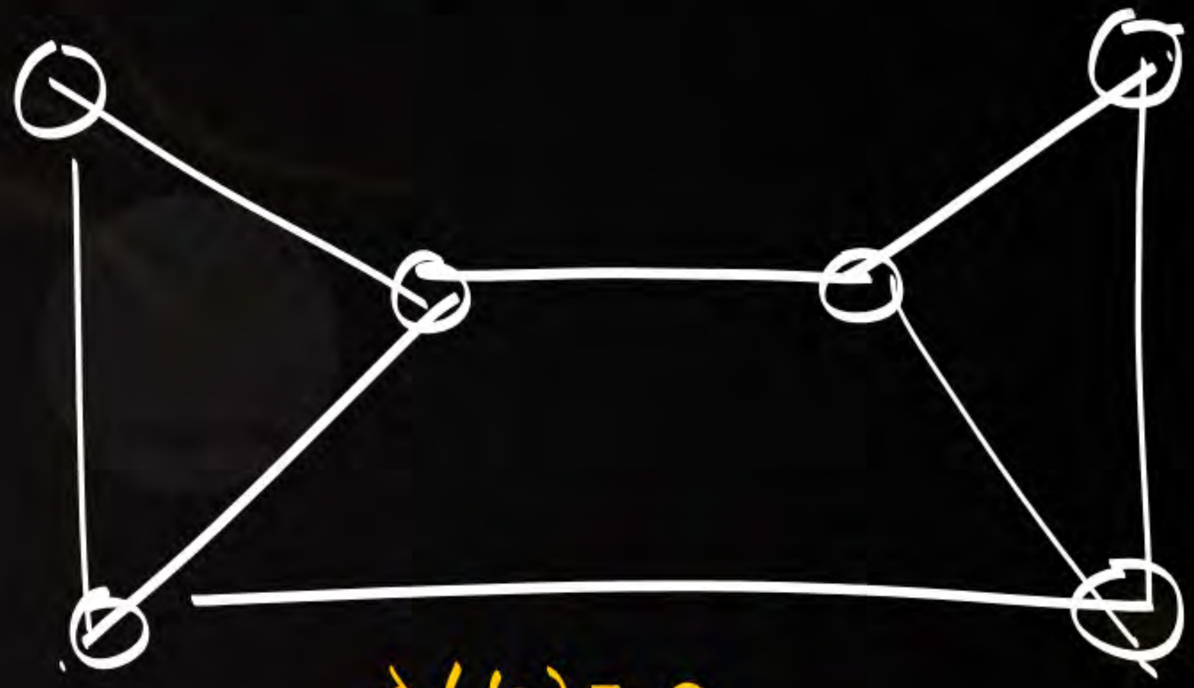
edge connectivity ($\lambda(G)$)

removal of min no of edges from a Graph will make graph as a disconnected Graph.

$\lambda(G) = 2.$



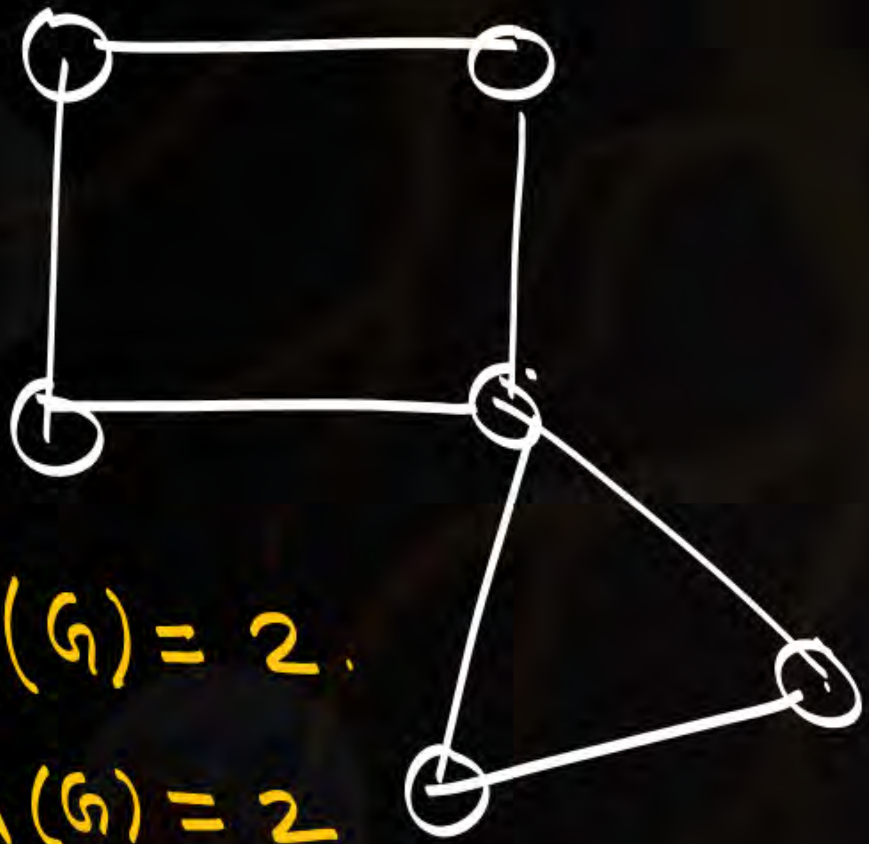
$\lambda(G) = 2.$



$\lambda(G) = 2.$

Every graph is having minimum degree.

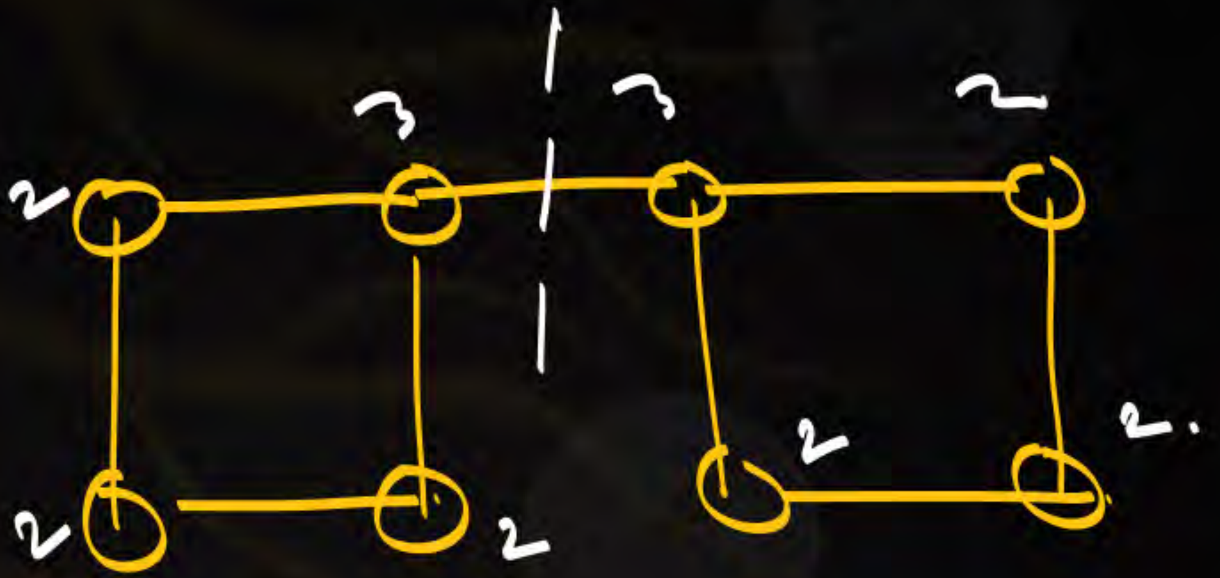
$\lambda(G)$ ~~min no. of edges associated with it.~~



$\delta(G) = 2$

$\lambda(G) = 2$

Case 1: $\lambda(G) = \delta(G)$



$\delta(G) = 2$

$\lambda(G) = 1$

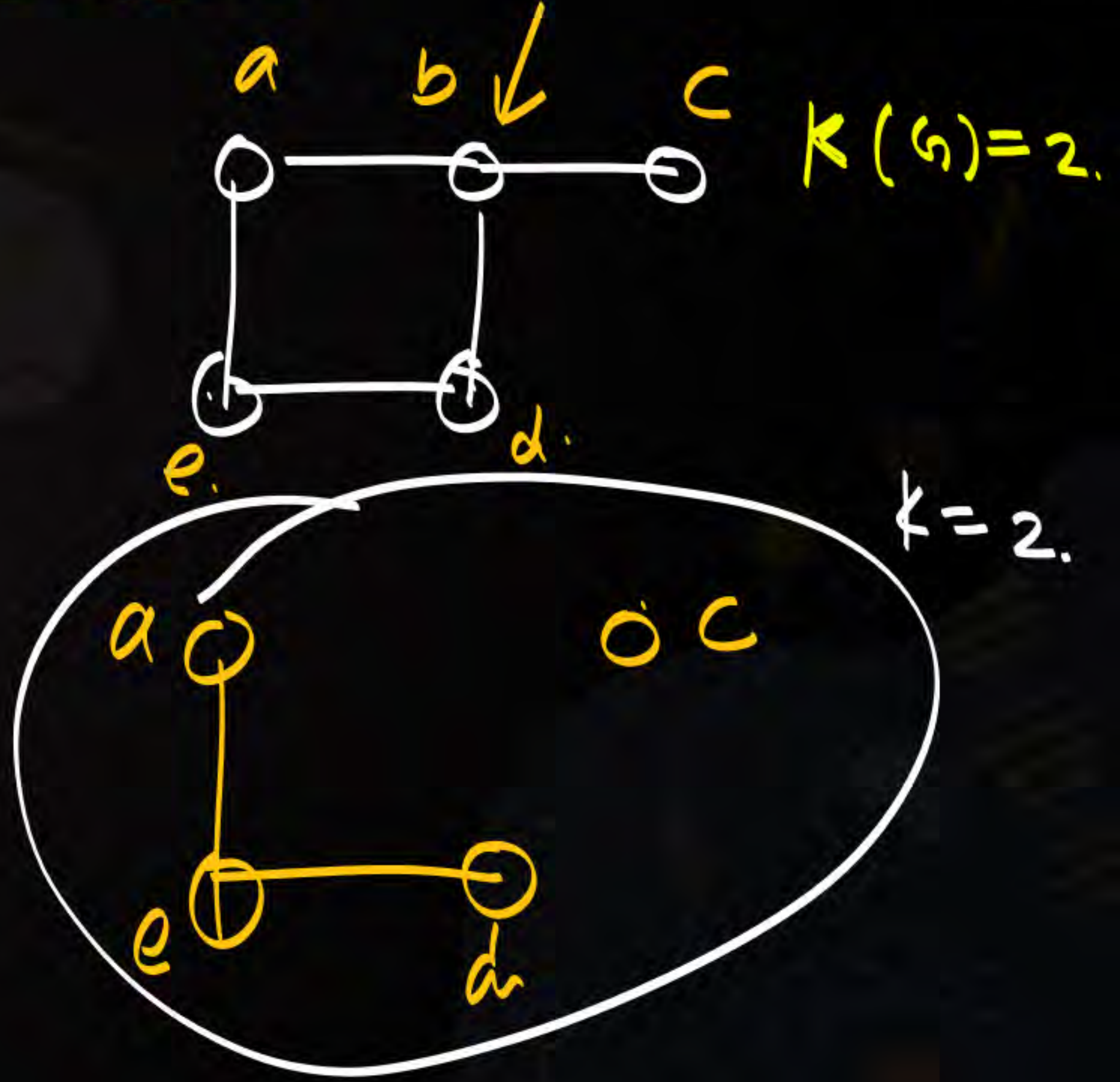
Case no: 2

$\lambda(G) < \delta(G)$

$$\boxed{\lambda(G) \leq \delta(G)}$$

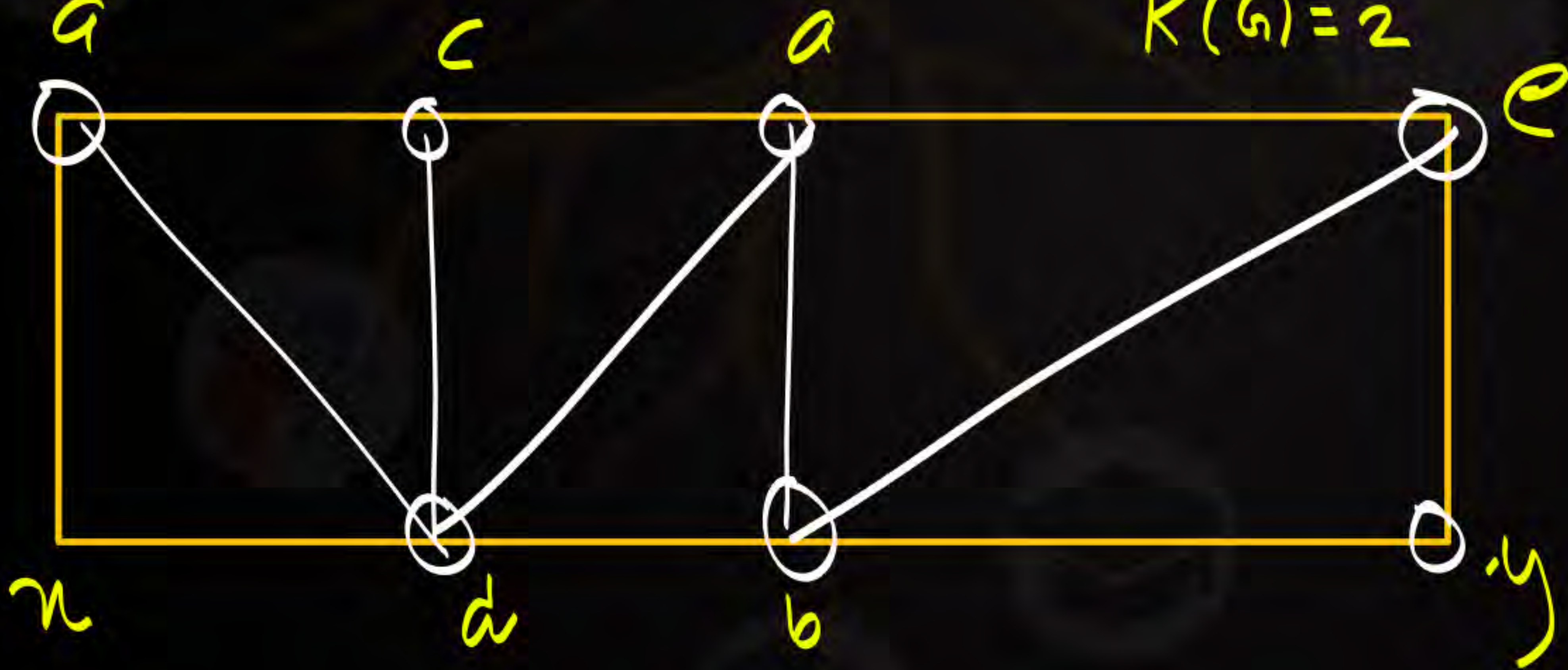
Cut vertex / Cut point / Articulation point.

Removal of single vertex from a graph will make graph as a disconnected graph.



Cut vertex set :

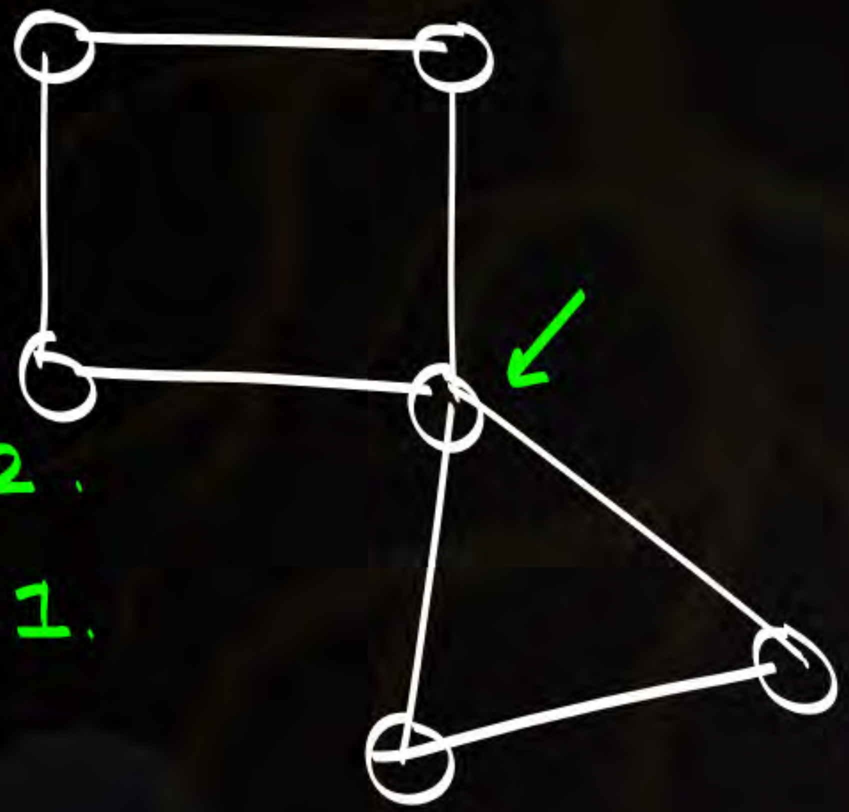
Removal of set of vertices from a graph will make graph as a disconnected graph.



vertex connectivity

$(K(G))$

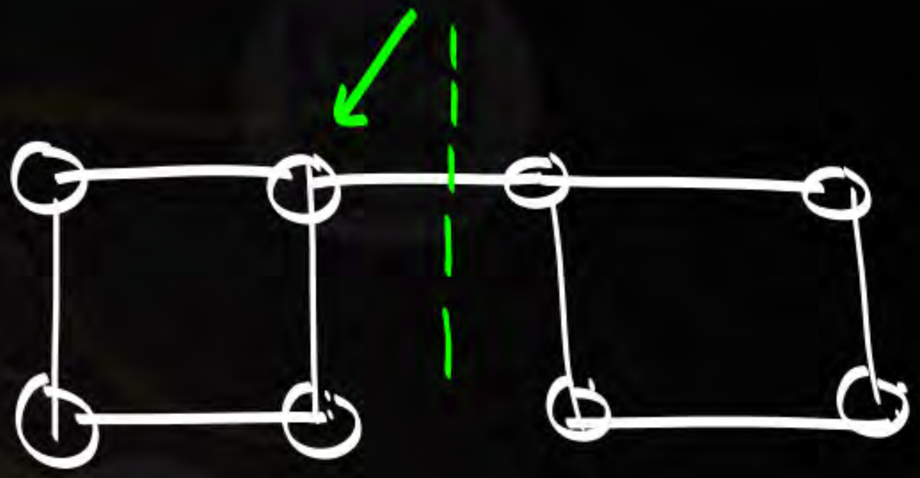
Removal of min no. of vertices from a graph will make graph as a disconnected graph.



$\lambda(G) = 2$

$K(G) = 1$

$K(G) < \lambda(G)$ — I



$K(G) = 1$

$\lambda(G) = 1$

$K(G) = \lambda(G)$ — II

$$\boxed{K(G) \leq \lambda(G)}$$

Thm: $\lambda(G) \le \delta(G)$

Thm: $K(G) \le \lambda(G)$

$K(G) \le \lambda(G) \le \delta(G)$

Thm 5: $\delta(G) \le \frac{2e}{n} \le \Delta(G) \le n-1.$

$$\boxed{K(G) \le \lambda(G) \le \delta(G) \le \frac{2e}{n} \le \Delta(G) \le n-1.}$$

if G is connected graph with 10 vertices & vertex connectivity 3, then minimum no. of edges necessary in G?

$K(G) = 3$. $n = 10$

$K(G) \leq \lambda(G) \leq \delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1$.

$K(G) \leq \frac{2e}{n}$

$3 \leq \frac{2e}{10}$

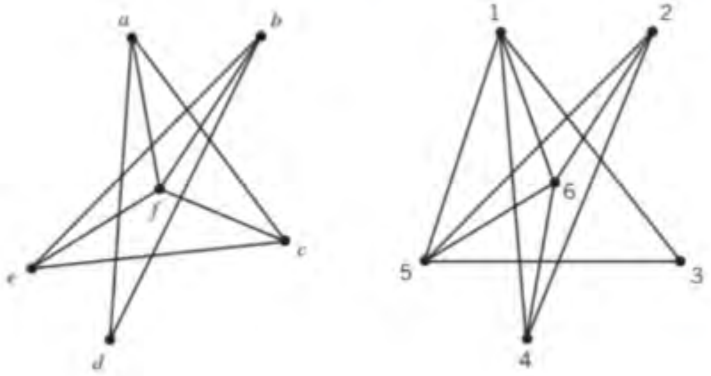
$\frac{30}{2} \leq e$

$15 \leq e$

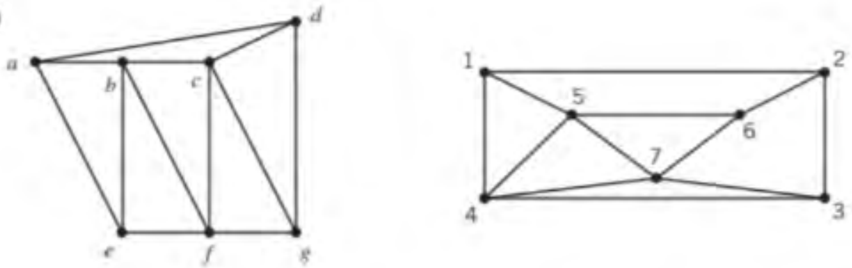
$15 \leq e$

$e = 15$

(e)

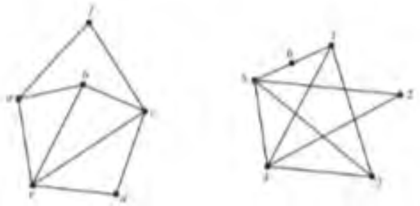


(f)



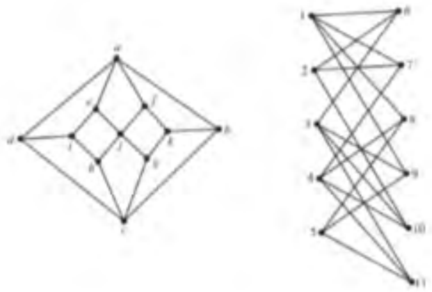
(g)

(i)

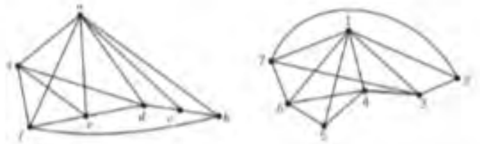


(j)

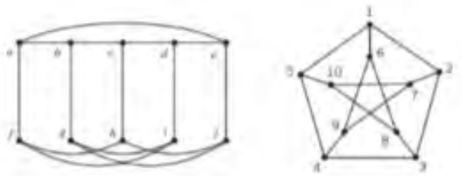
(k)



(h)



(l)



2 mins Summary

Topic One — Connectivity in Graph Part-03

Topic Two

Topic Three

Topic Four

Topic Five

THANK - YOU